

1. (5 pontos) Based on Kurlat (2020, Cap. 10 e 11)

For any given period, the nominal value of total purchases of goods and services is given by PY , and these transactions need to be paid with currency or deposits, so by definition it must be that

$$MV = PY \quad (1)$$

Where V accounts for how many times a unit of money has circulated in the economy over a given period, which is called the *velocity of circulation of money*.

- a) Why is this equation an accounting identity rather than, by itself, a theory of money demand? In particular, does it tell us whether a change in M causes a change in p , Y , or V ?
- b) Based on Equation 1 and the Baumol-Tobin model seen in class. What does this model imply about the relationship between money demand and the velocity of money?

If we define, in theory, that real money demand is given by $m^D(Y, i)$. Being an increasing function of real income Y and a decreasing function of the nominal interest rate i . Then, the money market equilibrium is given by:

$$M^S = p m^D(Y, i). \quad (2)$$

- c) If prices are fixed ($p = \bar{p}$) in the short-run, how must real income and nominal interest rates change after an exogenous increase in money supply? Is there a clear mechanism for how this change occurs? Do your answers change if prices were flexible?
 - d) How do we interpret the money market equilibrium if we assume Central Banks control the money supply? What if they control the nominal interest rate instead?
2. (5 pontos) Suppose the economy is in a steady state in which the money supply grows at rate μ , real output grows at rate g , and the nominal interest rate is constant, then we can denote the income elasticity of real money demand η by

$$\eta \equiv \frac{\partial m^D(Y, i)}{\partial Y} \frac{Y}{m^D(Y, i)} \quad (3)$$

- a) Use the money-market equilibrium (Equation 2) condition to show that, in this setting, the inflation rate is given by $\pi = \mu - \eta g$
- b) Assume a central bank targets $\pi^* = 2\%$ in an economy growing at $g = 3\%$. Based on this:
 - i. If money demand follows the Baumol-Tobin model, what is the value of η ?
 - ii. What money growth rate should the central bank choose in this case?
 - iii. If the central bank is wrong and money demand actually follows the Cambridge Equation (with $\eta = 1$), what will be the actual inflation rate?

- c) Now, assume that alongside output growth g , the fixed cost of a bank trip falls at a constant rate: $\frac{\dot{F}}{F} = f < 0$. Using the Baumol-Tobin demand and assuming a constant nominal interest rate i , show that:

$$\pi = \mu - \frac{1}{2}(g + f)$$

Also, explain why a falling F is inflationary under a constant money supply growth rule.